

Underline

Repo-based one-page summary generated from README, package manifests, and source code.

What It Is

Underline is a Chrome reading-tutor MVP built as a browser extension plus a local TypeScript API. It helps readers highlight confusing text on web articles and receive a short bridge explanation tailored to their background.

Who It's For

Primary persona: a non-technical or low-familiarity learner reading technical content. Evidence in repo: the default profile is "Non-technical learner" and the popup copy says the app infers missing background knowledge from highlighted phrases.

What It Does

- Chrome extension UI with popup, options page, and in-page reader mode controls.
- Captures selected text as structured highlights with DOM path, offsets, and quote context.
- Builds a small paragraph window around highlights instead of sending full-page HTML.
- Calls a local API for explanations and inserts an AI bridge paragraph back into the article.
- Stores highlights, bridges, reader settings, and lightweight learning signals in `chrome.storage.local`.
- Lets the user save provider URL/key/model, test the connection, and fall back to mock mode if upstream fails.

How It Works

- **Frontend:** `apps/extension` uses WXT + React for the popup/options UI and a content script that manages highlights and injected bridge paragraphs.
- **Backend:** `apps/api` runs a Fastify server with `/health`, `provider-config` routes, and `/v1/explanations`.
- **Shared contract:** `packages/shared` holds Zod schemas and shared types for requests, responses, highlights, bridges, and profile data.
- **Data flow:** user selects text -> extension stores highlight + context -> extension POSTs structured JSON to local API -> API calls OpenAI-compatible or mock client -> extension inserts returned bridge text after the anchor paragraph.
- **Persistence:** reading state lives in `chrome.storage.local`; provider runtime config is saved by the API to `.underline-runtime.json`.
- **Not found in repo:** database, cloud deployment, authentication, or server-side HTML scraping.

How To Run

- Install dependencies: `npm install`
- Start the local API: `npm run dev:api`
- Start the extension dev build: `npm run dev:extension`
- In Chrome, open `chrome://extensions`, enable Developer mode, and load `apps/extension/.output/chrome-mv3`.
- Open the popup, keep local API at `http://localhost:8787`, add provider URL/key/model if desired, test, then save.

Validation commands in repo: `npm run build`, `npm run test`, `npm run typecheck`.